

Agent-Based Modeling of the Ductal Structure in Pancreatic Precancer

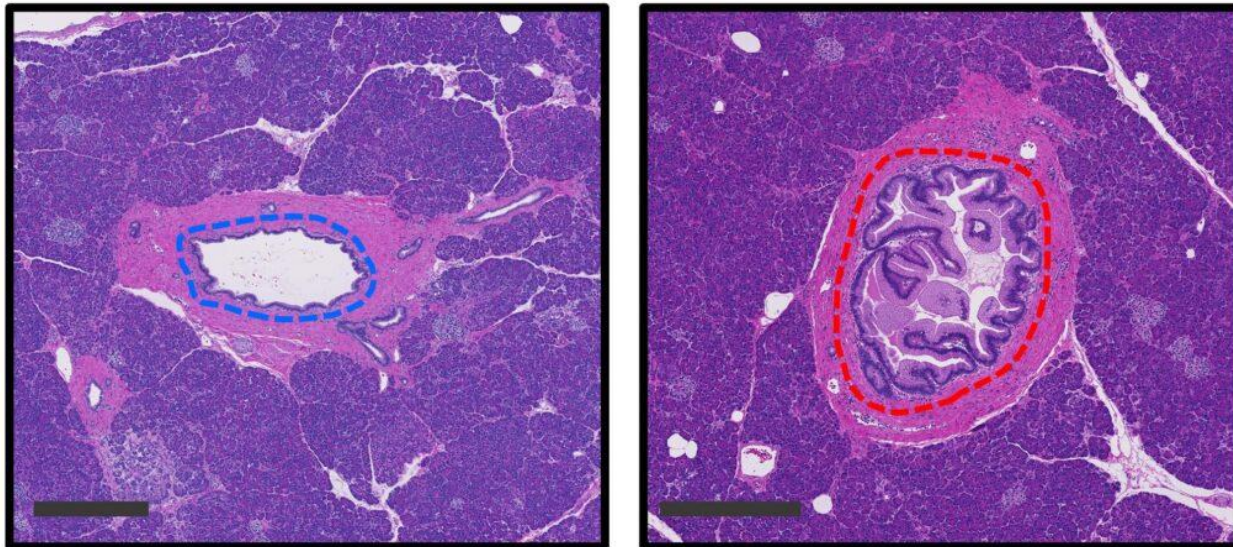
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How do PanINs Evolve?

Project Goal: Understand Coevolution of Pancreatic Ducts and Precancerous Cells

1. How do precancerous cells alter ductal structure?



Normal pancreatic duct



PanIN (precursor lesion)



UNIVERSITY of MARYLAND
SCHOOL OF MEDICINE

Kiemen A, Wu PH, Braxton AM, Cornish TC, Hruban RH, Wood LD, et al. Sci Adv. 2024
DOI: <https://doi.org/10.1126/sciadv.ado5103>.

Pancreatic Precursor Lesions (PanINs)

Project Goal: Understand Coevolution of Pancreatic Ducts and Precancerous Cells

2. How does evolving ductal structure affect cellular phenotypes?

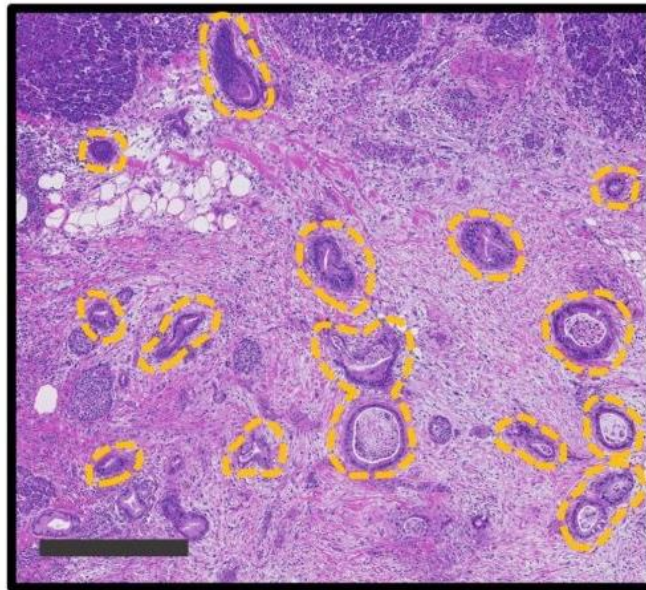


Cell Type Classification

Pancreatic Precursor Lesions (PanINs)

Project Goal: Understand Coevolution of Pancreatic Ducts and Precancerous Cells

3. How does this process shape progression and invasion

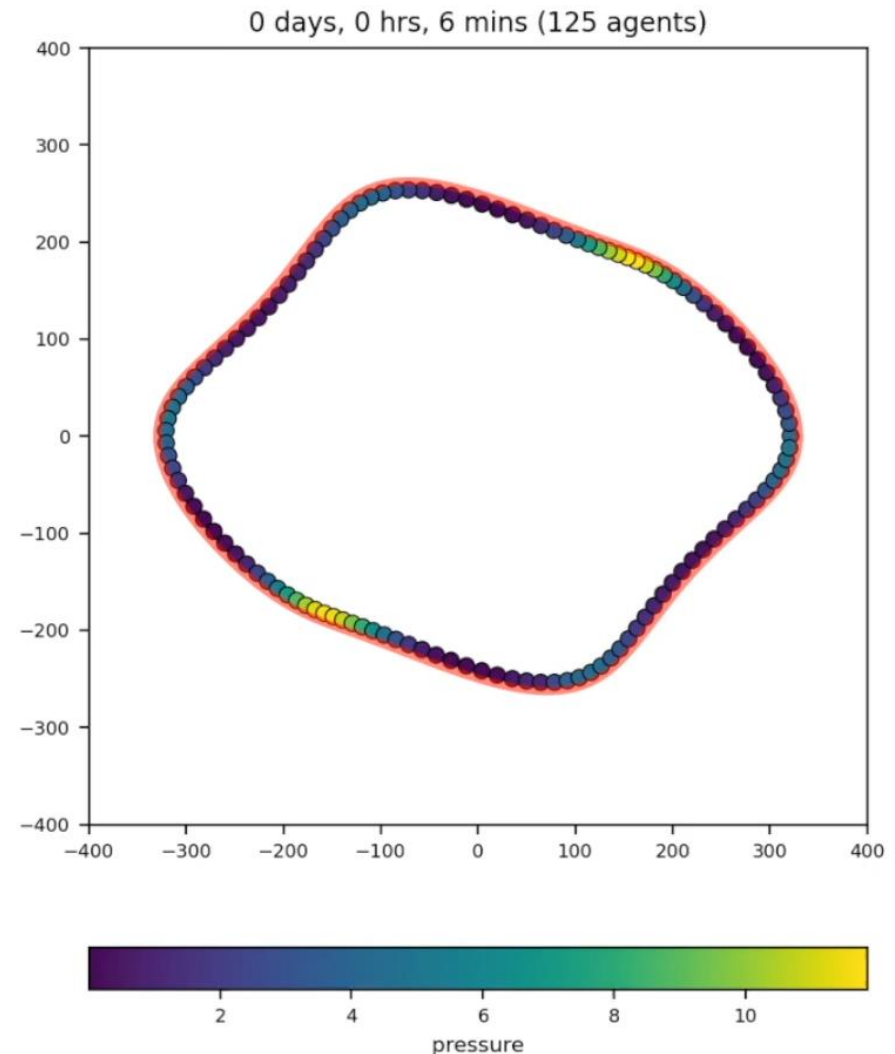


→ PDAC (invasive cancer)

Using Agent-Based Models to Simulate PanIN Evolution

Modeling Goals

1. Deformation of ductal structure by growing epithelial layer
2. Precancerous cells grow along the duct
3. Interaction with stromal cells
4. Acquisition of invasive phenotype

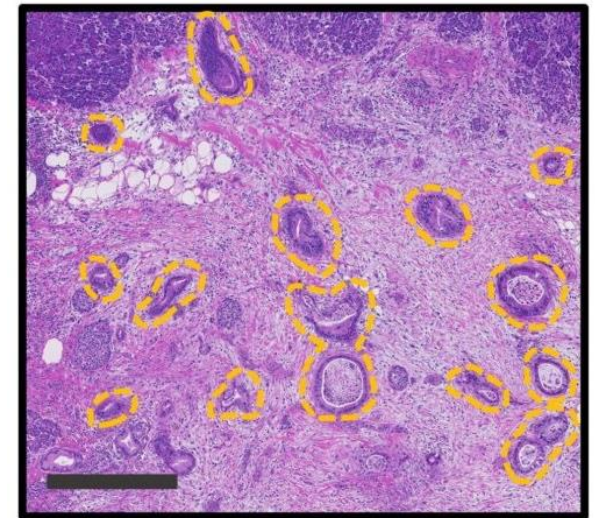
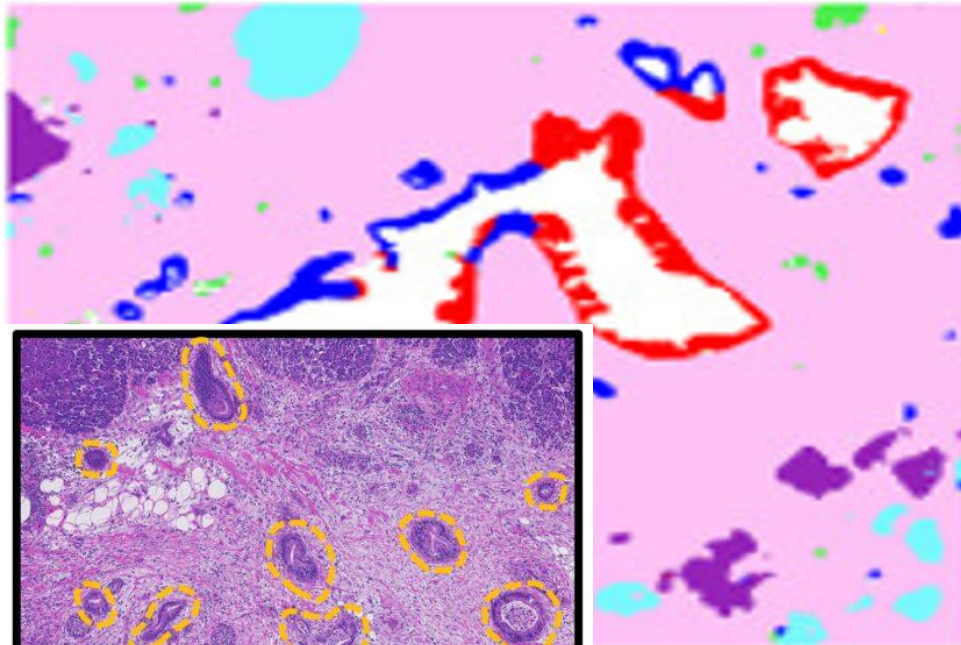


Bridge Simulated and Experimental Data

Aim: Bridge the gap between experimental data and simulated ductal structures

3. Quantify the morphological variance between simulated ductal structures and ST ground truths.
4. Calibrate parameters to identify true mechanistic drivers

Validation



→ PDAC (invasive cancer)